

LANESKIFF · REQUIREMENTS

LaneSkiff requirements

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DOCUMENT CONTROL

Document ID	LSK-REQ-001	Version	0.4
Status	Draft	Date	2026-10-04
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/laneskiff		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-04	Amish Chadha	R2 restated by Amish's round-3 decision 6A (LSK-DDR-004): payload 375 kg (five persons) within the R3 draft limit, ratio reported; met
0.3	2026-10-03	Amish Chadha	Amish's decisions 30A, 31C and 32A (LSK-DDR-003) carried out; rated load restated as five persons (375 kg); R5 states the stern quarter foam and the aft-crew rule; R10 left out of the first prototype; TRL 3 status from LSK-CAL-001 v0.2; numeric targets unchanged
0.2	2026-10-03	Amish Chadha	Rated load and test conditions stated; TRL 3 status from LSK-CAL-001; targets unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

LaneSkiff requirements

Requirements for the first prototype, built on the plywood and epoxy path in the light timber specification (okoume plywood, 4 mm sides, softwood framing). Numeric targets are those set at TRL 1 and are unchanged. On 2026-10-03 Amish decided the open requirement items (LSK-DDR-003): "i agree with all the 46 recommendations you provided. please proceed." Decision 30A sets the rated load at five persons, 31C adds stern quarter foam and the aft-crew rule to R5, and 32A leaves the fin drive out of the first prototype. Status is the paper result at TRL 3 from LSK-CAL-001; every requirement is verified by test at TRL 4. On 2026-10-04 Amish decided option 6A on R2: "For round 3, I agree with all your proposed recommendations". R2 is restated as "payload 375 kg (five persons) within the R3 draft limit; the payload-to-hull ratio is reported" (LSK-DDR-004). Where a requirement is not met or is at risk on paper, the state and the options are set out for Amish in docs/REVIEW.md and listed as open decisions in LSK-DEC-001.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 4)	TRL 3 status
R1	Carry mass	Complete hull 50 kg or less; each half 25 kg or less (plywood and epoxy path, target)	Weigh the prototype halves	Not met on paper: hull 87.9 kg, halves 49.9 and 38.1 kg; accepted by Amish with the light timber specification and shoulder slings (decision 30A)
R2	Payload	Payload 375 kg (five persons) within the R3 draft limit; the payload-to-hull ratio is reported, not a pass mark (restated 2026-10-04, decision 6A)	Static load test with water ballast in calm water	Met on paper as restated: 375 kg at 148 mm draft, inside the 150 mm limit. Reported ratio: 4.3 times hull mass at the rated load; 401 kg fits within the R3 draft (4.6)
R3	Draft at rated load	15 cm or less at the rated load of five persons (375 kg), Amish, 2026-10-03, decision 30A: "i agree with all the 46 recommendations you provided. please proceed."	Measured at the transom and bow during the load test	Met on paper: 148 mm at the transom, 138 mm at the forward end of the waterline
R4	Width for lanes	Beam 1.2 m or less overall	Tape measure; trial passage through a 1.3 m gate	Met: 1,197 mm over the rub strakes (3 mm margin after the 60 mm wider bottom)
R5	Level flotation when swamped	Floats approximately level with rated persons aboard when fully swamped, using the ABYC H-8 test method (not certified); 30 L of foam in the stern quarters, and the aft crew moves amidships when swamped (operating rule), Amish, 2026-10-03, decision 31C: "i agree with all the 46 recommendations you provided. please proceed."	Swamp test in a pool following ABYC H-8 procedure, with and without the aft-crew rule	Met on paper: 0.4 deg trim and 161 mm freeboard at the transom with the foam alone; 216 mm at the transom and 141 mm at the bow with the rule as well
R6	Boarding from waist-deep water	An 80 kg adult boards over the stern step unaided; heel stays under 10 degrees (target)	Boarding trials with several volunteers of different builds, inclinometer on the hull	Met on paper: 3.3 deg heel over the stern step (12.8 deg over the side for comparison)
R7	Assembly	Two halves joined in 10 min or less by two people with one spanner	Timed assembly trial	Met on paper (estimate): about 6 min, one 17 mm spanner
R8	Local build	At least two build paths (plywood and epoxy, aluminium or plastic sheet) documented with mass and payload	Build by a partner boatyard from the published drawings	Met on paper: three paths in the material substitution table (LSK-CAL-001, section 8)
R9	Prototype cost	USD 1,500 or less including step, poles and paddles	Bill of materials and receipts	USD 1,961: over the value-engineering target by USD 461
R10				

ID	Requirement	Target	Verification (TRL 4)	TRL 3 status
	Optional fin drive	Removable in 5 min or less; the boat meets R2 to R6 without it. Not part of the first prototype, Amish, 2026-10-03, decision 32A: "i agree with all the 46 recommendations you provided. please proceed."	Trials with and without the drive fitted, once a drive exists	Not shown on paper: left out of the first prototype by decision; R2 to R6 are assessed without a drive

■ Definitions used for the TRL 3 status

- **Hull mass:** the complete boat with its stern step, handles, strakes, skids and buoyancy, as carried; the push pole, paddles and bow line are loose gear (6.2 kg) and not counted.
- **Rated load:** five persons at 75 kg (a crew of two and three adults, 375 kg), placed as in LSK-CAL-001, section 3; decided by Amish on 2026-10-03 (decision 30A) and marked on the transom.
- **Swamped:** both halves full to the outside water level, persons aboard counted at two thirds of their weight (a conservative design assumption, not the ABYC H-8 rule itself).
- **Draft:** from the water surface to the outside of the bottom, at the transom and at the forward end of the waterline.

■ Assumptions

- Fresh floodwater, 1,000 kg/m³; calm, slow water.
- Okoume marine plywood of about 450 kg/m³ and softwood framing of about 500 kg/m³ (light timber specification, decision 30A); hardwood of about 700 kg/m³ only for the step, the bow eye pad and the rung.
- Closed-cell polyethylene foam of about 30 kg/m³ with under 1 % water uptake by volume is available near the first partner.
- Two people can carry about 40 to 50 kg between them over a short distance; four can carry the assembled boat.
- A stern-centre step heels the boat less than a side step; LSK-CAL-001 confirms this on paper, the boarding trials confirm it in water.

Safety: LaneSkiff is an open engineering reference for a rescue boat, not certified equipment. Every trial behind R2, R3, R5 and R6 is run in calm, shallow water with a safety boat or bank team, every person in a life jacket, and the volunteers briefed; no trial uses moving water. The step hinge and stop straps are proof-loaded to twice their working load before any boarding trial. The boat carries no more than five persons; when swamped, the aft crew moves amidships and everyone stays seated, low and centred.